

Soil Health & Crop Recommendation Report

Report Date: November 06, 2025 at 20:57:49 UTC

Farmer Information	
Name:	sureshm bajan
Phone:	9482601807
Location:	pune
Land Size:	23.00 acres

Parameter	Value	Unit
Nitrogen (N)	85.0	mg/kg
Phosphorus (P)	93.0	mg/kg
Potassium (K)	80.0	mg/kg
pH Level	6.50	
Temperature	31.6	°C
Humidity	70.4	%
Rainfall	250.0	mm
Soil Type	Alluvial Soil	

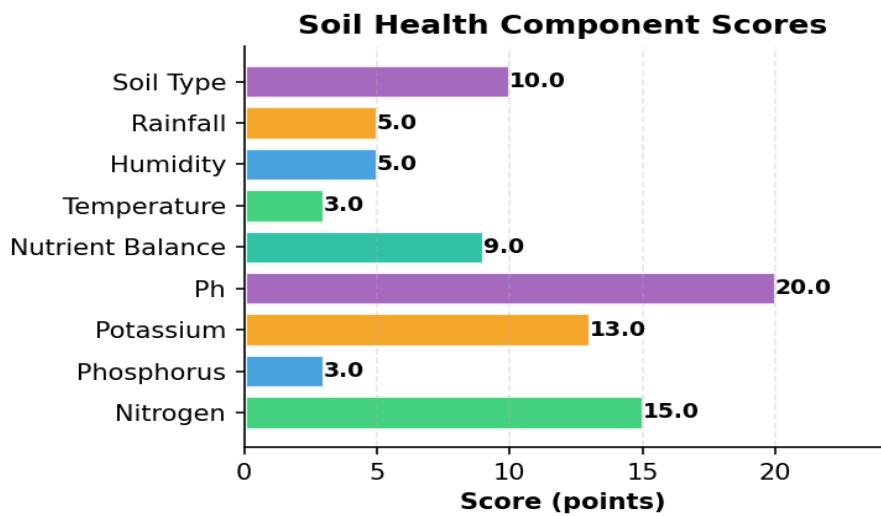
Soil Health Assessment

Overall Soil Health Score	83.0/100	Excellent
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Component	Score	Points
Nitrogen	15.0	points
Phosphorus	3.0	points
Potassium	13.0	points
Ph	20.0	points
Nutrient Balance	9.0	points
Temperature	3.0	points
Humidity	5.0	points
Rainfall	5.0	points
Soil Type	10.0	points

Soil Health & Crop Recommendation System

Comprehensive Agricultural Analysis Report



Recommended Crop

Predicted Crop	Rice
Confidence Level	50.5%

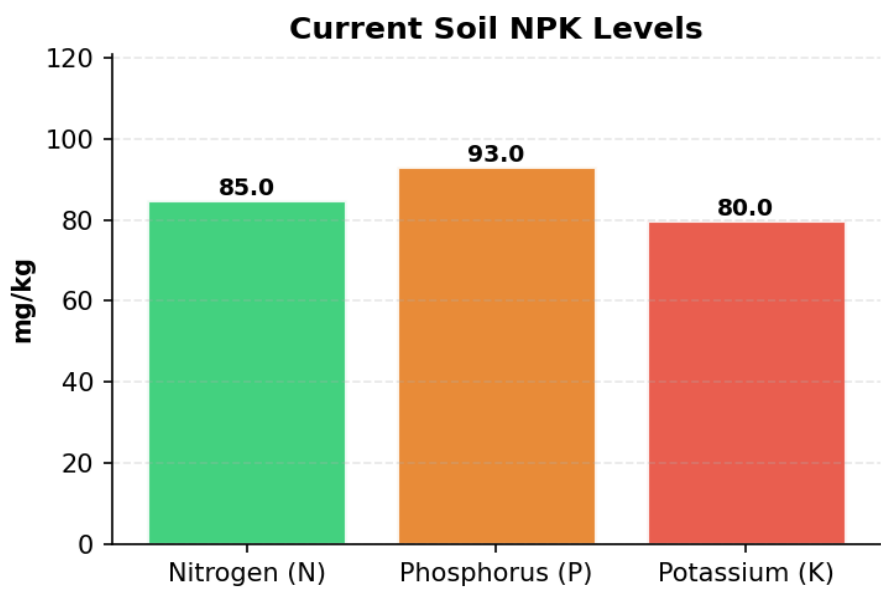
Alternative Crop Options

Rank	Crop	Confidence (%)
1	Banana	16.6%
2	Jute	5.6%

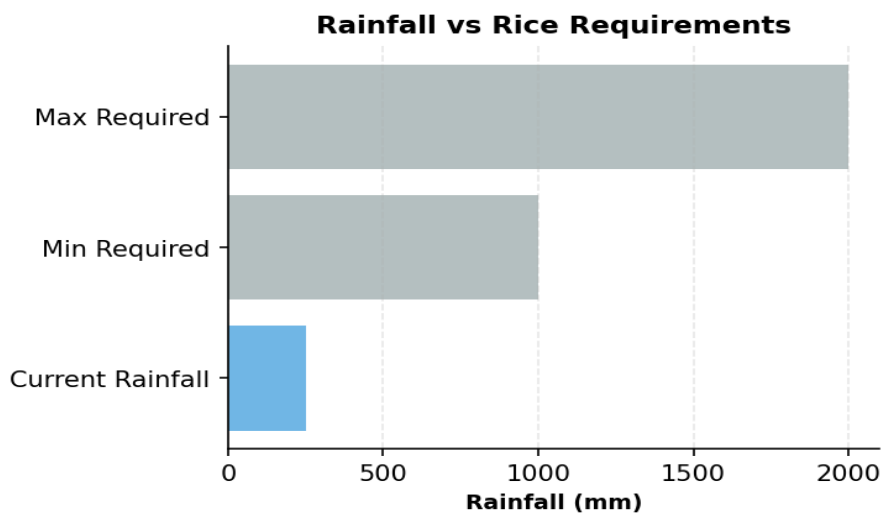
Soil NPK Levels Visualization

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Rainfall Analysis



Important Warnings

- Very high phosphorus levels detected. May interfere with micronutrient uptake.

Soil Treatment Recommendations

- Soil pH (6.5) is within optimal range (5.5-7.0) for rice.
- Nitrogen (85.0 mg/kg) is within optimal range for rice.
- Phosphorus (93.0 mg/kg) is excessive for rice. Optimal range: 30-50. High P can interfere with micronutrient uptake.
- Potassium (80.0 mg/kg) is excessive for rice. Optimal range: 30-50. High K can cause magnesium deficiency.

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- Rich in nutrients, good water retention. Suitable for most crops.

Fertilizer Recommendations

■ ****Fertilizer Requirements for rice:****

- Land Size: 23.0 acres
- Current NPK: N=85.0, P=93.0, K=80.0 mg/kg
- Ideal NPK: N=90.0, P=40.0, K=40.0 kg/acre
- ****Required Application:****
- Nitrogen (N): 115.0 kg total (5.0 kg/acre)
- → Apply in 2.0-3.0 split doses: 50.0% at sowing, 25.0% at tillering, 25.0% at flowering
- ■ ****Organic Alternatives:****
- Compost/FYM: 5.0-10.0 tons/acre (provides N, P, K)
- Green Manure: Leguminous crops (fixes N)
- Biofertilizers: Rhizobium (N), PSB (P), VAM (P)

Crop-Specific Tips for Rice

Ensure consistent water supply and transplant in puddled field. Maintain 5-7 cm water depth.

Additional Notes

- These recommendations are based on comprehensive soil analysis and advanced ML predictions.
- Actual results may vary based on local conditions, weather patterns, and farming practices.
- Consult with local agricultural experts before making major farming decisions.
- Regular soil testing (every 2-3 years) is recommended for optimal crop management.
- Consider crop rotation and organic farming practices for long-term soil health.
- Monitor weather forecasts and adjust irrigation schedules accordingly.